NGOWA KARISA

MECHANICAL ENGINEER · FULL-STACK DEVELOPER

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◆ DETAILS

EMAIL
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LINKEDIN
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◆ SKILLS

FRONTEND

React 18/19 · TypeScript · Vite · Tailwind CSS · Framer Motion · Three.js · React Query v5 · Zod · Vitest

BACKEND & DB

Node.js · Express.js · Supabase · PostgreSQL · RLS / RBAC · Clerk Auth · Cloudinary

DEVOPS

Vercel · Heroku · AWS · GitHub Actions · CI/CD

ENGINEERING

MATLAB · CATIA · AutoCAD · FEA · WebGL

◆ APPROACH

Systems thinking from mechanical engineering applied to software architecture. Performance-

◆ PROFILE

Full-Stack Developer with a foundation in Mechanical Engineering — bringing systems thinking, precision, and a performance-obsessed mindset to every product I build. I architect production-grade web applications end-to-end: from PostgreSQL schemas and Supabase backends to React 19 frontends with sub-1.2s load times, 90%+ test coverage, and enterprise RBAC systems. Three years shipping paid client software that scales; background in CAD and MATLAB-driven computational analysis. I don't just write code — I engineer solutions.

◆ EMPLOYMENT HISTORY

Full-Stack Developer — Freelance
2022 — Present

Independent / Client Contracts

- Designed and delivered end-to-end production platforms for real estate, non-profit, and engineering clients — sole developer from schema design to deployment.
- Architected PostgreSQL schemas with full indexing strategies and row-level security policies.
- Implemented RBAC systems with granular permission matrices (up to 20+ permission types, 6 roles).
- Achieved sub-1.2s load times and 95/100 Lighthouse mobile scores across all delivered projects.
- Established 90% unit/integration test coverage using Vitest + React Testing Library.
- Integrated third-party services: Clerk, Cloudinary, EmailJS, Vercel Analytics, Sentry.
- Maintained 99.9% uptime on all production systems via automated CI/CD (GitHub Actions + Vercel).

Full-Stack Developer — Raslipwani Properties
2023 — 2024

Client Contract · raslipwani.co.ke

React 18 · Vite · Supabase · PostgreSQL · React Query · Clerk Auth · Cloudinary · Vercel · Vitest

- Built an enterprise property management SaaS platform from zero — booking engine, full CRM, admin analytics dashboard — supporting 100+ concurrent users.
- Reduced page load time from 3 s → **1.2 s** (60% improvement) via React Query caching, server-side pagination (95% data reduction), and debounced search (80% fewer API calls).
- Engineered a complete CRM system — client lifecycle, property interest tracking, communication timelines, and internal notes — delivered in a single sprint.
- Designed a 6-table relational schema with 40+ indexes across properties, bookings, clients, interests, communications, and admin_settings.

obsessed. Ships tested, documented, zero-downtime production code.

- Achieved **90% test coverage** across 27 files / 5,700+ lines of code; Mobile Lighthouse **95/100**, Desktop **98/100**.
- Built complex booking workflow: drag-and-drop rescheduling, multi-status pipelines (pending → confirmed → completed), automated email triggers.

Full-Stack Developer — Neema Foundation Kilifi
2024 — 2025

Client Contract · neemafoundationkilifi.org

React 19 · TypeScript 5.9 · Vite 7 · Supabase · Three.js · TipTap · DND Kit · Zod · Framer Motion 12 · Vercel

- Delivered a full digital transformation platform for a Kenyan non-profit managing healthcare, education, and community empowerment programs.
- Architected a **5-tier RBAC system** (Super Admin → Viewer) from scratch with 20+ granular permission controls across 7 admin CMS pages.
- Implemented DOMPurify XSS protection across all rich-text editor inputs; achieved Lighthouse Accessibility **95+**, SEO **95+**.
- Optimised LCP to **<2.5 s** on 3G Fast; CLS <0.1; TBT <200 ms.
- Integrated Three.js animated hero with graceful prefers-reduced-motion fallback; built 10-table PostgreSQL schema with Supabase RLS.
- Created a comprehensive TypeScript strict-mode type system with ESLint 9 enforcement; zero runtime type errors in production.

Frontend / WebGL Developer — CAD Web Viewer
2022 — 2023

Commissioned Engineering Contract

Three.js · React · Custom WebGL Shaders

- Designed and built a browser-based 3D model viewer for mechanical engineering components, eliminating the need for desktop CAD software in client review cycles.
- Reduced client review cycles by **65%** by enabling direct browser-based model inspection and markup without installed software.
- Authored custom WebGL shaders for real-time material and lighting simulation of engineering-grade components.

◆ EDUCATION

B.Sc — Computer Information Systems
2024 — Present

Kenya Methodist University

Currently enrolled. Relevant modules: Software Engineering · Database Systems · Systems Analysis & Design · Computer Networks · Web Technologies

B.Eng — Mechanical Engineering
2019 — 2023

Shenyang Agricultural University

Thesis: "Optimization of Mechanical Systems Using Computational Methods"

Relevant coursework: Computational Methods · Finite Element Analysis · Thermodynamics · Systems Engineering · MATLAB-driven Data Analysis · CATIA 3D Modelling · Structural Analysis

◆ REFERENCES

> *"I apply mechanical precision to digital solutions — building web applications with the same rigour I'd engineer physical systems. Performance optimisation isn't just*

> *about faster code; it's about creating efficient, maintainable systems that withstand real-world demands."*

- Full references and client testimonials available upon request.